

Support Geometry and Nearest Neighbours:

Does the Tangential–Orthogonal–Depth Decomposition Add Information
Beyond Feature Lines, Local Hyperplane Distances and Depth?

Luciano Silva Alarco

Pontificia Universidad Católica del Perú

ORCID 0000-0002-3395-3129

Working draft v0.3 (two rounds of internal review applied) — research line SPD001 — 3 October 2026

Abstract

For a query q and its k nearest training neighbours of a class, the residual from the neighbourhood centroid splits into a *tangential* component along the local principal subspace and an *orthogonal* component normal to it, and the position of q among the neighbours can be summarised by a *depth*. The research line SPD001 asked whether this description carries information for classification beyond nearest feature lines, local hulls and depth measures. We note that the regularised local-hyperplane distance of Vincent and Bengio is, by the ridge identity, exactly a data-weighted sum of the orthogonal component and the squared tangential coordinates, $h_\lambda^2 = O^2 + \sum_j \lambda t_j^2 / (s_j^2 + \lambda)$, while the local spatial depth is not a function of the two distances. We run the isolating comparison the line had left pending (8 datasets, 5×5 repeated cross-validation, six tuned references, a conditional-logit combination of the three components and its ablations) under a criterion fixed before the first complete run, a paired bootstrap interval above zero against the best reference on at least 5 of 8 datasets, with hyper-parameter grids widened and preregistered after an internal review found the earlier ones truncating. The combination met the criterion on 0; its difference to the best reference ranged from -0.8 to $+0.3$ accuracy points, and the largest upper limit is $+1.4$ points with the bootstrap and $+3.9$ with the Nadeau–Bengio variance, so the study excludes gains of a few points, not of one. Removing the orthogonal component costs up to $+5.1$ points (interval above zero on 5 of 8 datasets with the bootstrap, 2 with Nadeau–Bengio; two effects seen with the narrower grids did not survive); removing the depth or the tangential component never reduces accuracy significantly. Where the selected neighbourhood is larger than the dimension, HKNN has no hull term and is a regularised local Mahalanobis distance; the decomposition does not add predictive information beyond these HKNN-type local distances.

1 Introduction

Nearest-neighbour classification [6, 11] uses only distances from the query to individual training points. A family of extensions replaces the point by a local set built from the query’s neighbours in each class: the line through two prototypes (nearest feature line, 18), the affine hull of the k nearest same-class neighbours with a ridge penalty (HKNN, 28), bounded hulls and hyperdisks [3], local principal subspaces [15] or locally adapted metrics [14]. All of them measure, in one way or another, how far the query lies *off* the local class support. A separate tradition, statistical depth [19, 26, 30], measures how central a point is *inside* a cloud, and depth-based classifiers assign the query to the class in which it is deepest [12, 17], with local versions for multimodal classes [22].

The research line SPD001 (“support geometry and nearest neighbours”) set out from the observation that these quantities are components of one decomposition: if μ is the centroid of the k nearest neighbours of q in class c and T_m the m -dimensional principal subspace of that neighbourhood, $q - \mu$ splits orthogonally into a tangential part in T_m and an orthogonal part normal to it, and a depth measures where q falls relative to the neighbours. The line’s public record states its objective as *determining whether this decomposition adds information beyond feature lines, hulls and depth measures*, and notes that the isolating comparison was pending. This paper runs it and writes down the answer, with the elementary facts that explain it.

Contributions. (i) Precise definitions of the tangential, orthogonal and depth components of a query with respect to a class neighbourhood (Section 2). (ii) Elementary propositions (Section 3): the regularised

HKNN distance is a weighted combination of the orthogonal component and the tangential coordinates, between the hull and the centroid distance; the restricted feature-line distance lies between the hull and the nearest-neighbour distance; the local spatial depth is not a function of the two distances, so it *could* carry information. (iii) The isolating comparison with a criterion stated before the first complete run and grids preregistered before the final run (Section 4), and its outcome (Section 5), which is negative under that criterion. (iv) A list of claims with their status (Section 6).

Scope. The paper is a methodological study on small public and synthetic datasets; it makes no claim about any application domain. “Adds information” is given one operational meaning (Section 4); other meanings are not tested and not denied. Every number comes from the scripts of Appendix A with a fixed seed.

2 Definitions

Let $(x_i, y_i)_{i=1}^n$ be a training set with $x_i \in \mathbb{R}^d$ and labels $y_i \in \{1, \dots, C\}$, standardised coordinate-wise with training statistics. Fix a query $q \in \mathbb{R}^d$, a class c and an integer k with $k \leq n_c$, where n_c is the number of training points of class c (when q is itself a training point it is excluded from its own neighbourhoods).

Definition 2.1 (class neighbourhood, centred SVD). $N_k^c(q) \subset \{x_i : y_i = c\}$ is the set of the k training points of class c nearest to q in Euclidean distance (ties broken deterministically). Let $\mu = \mu_c(q)$ be its mean, $V \in \mathbb{R}^{k \times d}$ the matrix with rows $x_i - \mu$, $x_i \in N_k^c(q)$, and $V = W \text{diag}(s_1, \dots, s_p) U^\top$ a thin singular value decomposition with $p = \min(k, d)$, $s_1 \geq \dots \geq s_p \geq 0$ and orthonormal columns $u_1, \dots, u_p$ of U . Write $r = q - \mu$ and $t_j = u_j^\top r$. Since the rows of V sum to zero, $\text{rank } V \leq \min(k - 1, d)$; we call the neighbourhood *generic* when equality holds, and write $\rho = \min(k - 1, d)$.

Definition 2.2 (tangential and orthogonal components). For $1 \leq m \leq \rho$ let $T_m = \text{span}(u_1, \dots, u_m)$ (the local principal, or tangent, subspace of dimension m) and P_m the orthogonal projector onto it. The *tangential distance* and the *orthogonal distance* of q to the class- c neighbourhood are

$$T_m(q, c) = \|P_m r\| = \left(\sum_{j \leq m} t_j^2 \right)^{1/2}, \quad O_m(q, c) = \|(I - P_m)r\| = \left(\|r\|^2 - \sum_{j \leq m} t_j^2 \right)^{1/2}.$$

Definition 2.3 (local hull, HKNN and feature-line distances). The *local hull distance* is $\text{dist}(q, \text{aff } N_k^c(q))$, the Euclidean distance from q to the affine hull of the neighbourhood. For $\lambda \geq 0$ the *penalised local-hyperplane distance* of Vincent and Bengio [28] is

$$h_\lambda(q, c)^2 = \min_{\alpha \in \mathbb{R}^k} \|r - V^\top \alpha\|^2 + \lambda \|\alpha\|^2.$$

We take as the HKNN score the value of this penalised objective; this is our reading of Section 3 of Vincent and Bengio [28]. If one takes instead the plain distance $\|r - V^\top \alpha^*\|$ at the penalised minimiser α^* , Proposition 3.3 and Remark 3.4 hold with weights $\lambda^2/(s_j^2 + \lambda)^2$ in place of $\lambda/(s_j^2 + \lambda)$; the empirical results are for the reading adopted here (we could not check Section 3 of the original against a copy). The *local feature-line distance* is $\ell(q, c) = \min_{x_i \neq x_j \in N_k^c(q)} \text{dist}(q, x_i + \mathbb{R}(x_j - x_i))$, the nearest-feature-line distance of Li and Lu [18] restricted to the k neighbours. The *local class distance* is $\bar{d}(q, c) = \frac{1}{k} \sum_{x_i \in N_k^c(q)} \|q - x_i\|$.

Definition 2.4 (local spatial depth). With the spatial sign $S(v) = v/\|v\|$ for $v \neq 0$ and $S(0) = 0$, the *local spatial depth* of q with respect to the class- c neighbourhood is

$$D(q, c) = 1 - \left\| \frac{1}{k} \sum_{x_i \in N_k^c(q)} S(x_i - q) \right\| \in [0, 1],$$

the spatial (L_1) depth of Vardi and Zhang [27] and Serfling [24], computed on the neighbourhood instead of on the whole class. We call $E = 1 - D$ the *outlyingness*.

The spatial depth equals 1 at the spatial median of the neighbourhood when the median is not itself a neighbour (at a neighbour x_i the term $S(0) = 0$ drops out and $D(x_i)$ may be below 1); the median is unique when the neighbours are not collinear [20]; D tends to 0 as $\|q\| \rightarrow \infty$ and is invariant under orthogonal transformations and translations but not under general affine maps [4, 24, 27] [literature]. Restricting a depth to a neighbourhood of the query is a simple proxy for the local depths of Paindaveine and Van Bever [22]; we do not use their symmetrised construction.

Definition 2.5 (classifiers). All classifiers below assign q to the class that minimises a score $\sigma(q, c)$ (or maximises the depth), with no class prior, except kNN, which takes a majority vote among the k_{NN} nearest neighbours of all classes (ties towards the class of the nearest neighbour).

- NFL: $\sigma = \ell(q, c)$; HKNN: $\sigma = h_\lambda(q, c)$; LPH (local PCA hull): $\sigma = O_m(q, c)$;
- LCD: $\sigma = \bar{d}(q, c)$; SD: $\sigma = E(q, c)$; T: $\sigma = T_m(q, c)$;
- TOD (the decomposition classifier): with the feature vector $f_c(q) = (\log T_m, \log O_m, \log E)(q, c)$, standardised feature-wise with training statistics, and a weight vector $w \in \mathbb{R}^3$ shared by all classes,

$$P(c | q) = \frac{\exp(-w^\top f_c(q))}{\sum_{c'} \exp(-w^\top f_{c'}(q))},$$

a conditional logit without intercepts; w is fitted by maximum likelihood with an L_2 penalty $0.001 \|w\|^2$ on leave-one-out features of the training fold. TO, TD, OD are the same model with one feature removed. (A one-feature model with a positive weight reproduces the corresponding argmin rule, so LPH and SD are the one-component ablations for O and for D .)

The logarithms make the combination a product of powers $T_m^{w_1} O_m^{w_2} E^{w_3}$, invariant to the scale of each feature; a floor of 10^{-6} is added inside each logarithm.

3 Elementary results

Proposition 3.1 (Pythagoras and the hull). *For every $m \leq \rho$, $\|q - \mu\|^2 = T_m^2 + O_m^2$. If the neighbourhood is generic then $O_\rho(q, c) = \text{dist}(q, \text{aff } N_k^c(q)) = h_0(q, c)$.*

Proof. The first identity is orthogonality of P_m and $I - P_m$. For the second, $\text{aff } N_k^c(q) = \mu + \text{row}(V) = \mu + \text{span}(u_j : s_j > 0)$; in the generic case this is $\mu + T_\rho$, so $\text{dist}(q, \text{aff } N_k^c(q)) = \|(I - P_\rho)r\| = O_\rho$. Finally $h_0^2 = \min_\alpha \|r - V^\top \alpha\|^2$ is the squared distance from r to $\text{row}(V)$, the same quantity. $\square$

Remark 3.2 (low dimension). When $k > d$ and the neighbourhood is generic, $\rho = d$, $\text{aff } N_k^c(q) = \mathbb{R}^d$ and the hull distance $h_0 = O_\rho$ vanishes identically. With $k \geq 5$ this holds for every k in our grids on the 4 datasets with $d < 5$ (iris, swiss roll, moons, spheres), and, at the neighbourhood size actually selected, for HKNN in all folds on 5 datasets (iris, swiss roll, moons, moons+10 noise, spheres) and in some folds on wine (80%), breast cancer (12%); only on digits does it always keep a hull term. There HKNN is the penalised tangential term of Proposition 3.3 below, $r^\top (I + V^\top V/\lambda)^{-1} r$, a regularised local Mahalanobis distance to the neighbourhood centroid that is zero for $\lambda = 0$, not a distance to a hull; LPH is non-trivial only because $m \leq d - 1$. “Hull” carries information only when $d \geq k$.

Proposition 3.3 (HKNN is a weighted tangential–orthogonal combination). *For a generic neighbourhood and every $\lambda > 0$,*

$$h_\lambda(q, c)^2 = O_\rho(q, c)^2 + \sum_{j=1}^{\rho} \frac{\lambda}{s_j^2 + \lambda} t_j^2.$$

Consequently $\lambda \mapsto h_\lambda$ is non-decreasing, $O_\rho \leq h_\lambda \leq \|q - \mu\|$, $h_\lambda \rightarrow O_\rho$ as $\lambda \downarrow 0$ and $h_\lambda \rightarrow \|q - \mu\|$ as $\lambda \rightarrow \infty$.

Proof. Write $V^\top \alpha = U \text{diag}(s) W^\top \alpha$. Decompose $\alpha = W\beta + \gamma$ with $W^\top \gamma = 0$, so that $V^\top \gamma = 0$ and $\|\alpha\|^2 = \|\beta\|^2 + \|\gamma\|^2$, and decompose $r = \sum_j t_j u_j + r_\perp$ with $r_\perp \perp \text{span}(U)$. Then

$$\|r - V^\top \alpha\|^2 + \lambda \|\alpha\|^2 = \sum_{j \leq \rho} (t_j - s_j \beta_j)^2 + \|r_\perp\|^2 + \lambda \|\beta\|^2 + \lambda \|\gamma\|^2.$$

The minimum has $\gamma = 0$ and, for each j , $\beta_j = s_j t_j / (s_j^2 + \lambda)$, with minimal value $(t_j - s_j \beta_j)^2 + \lambda \beta_j^2 = t_j^2 (\lambda^2 + \lambda s_j^2) / (s_j^2 + \lambda)^2 = \lambda t_j^2 / (s_j^2 + \lambda)$. For $j > \rho$ one has $s_j = 0$ in the generic case and the term equals t_j^2 ; together with $\|r_\perp\|^2$ these terms sum to $\|r\|^2 - \sum_{j \leq \rho} t_j^2 = O_\rho^2$ by Proposition 3.1. Each weight $\lambda / (s_j^2 + \lambda)$ lies in $(0, 1]$, increases with λ and tends to 0 or 1 at the two ends, which gives the monotonicity and the limits. $\square$

Proposition 3.3 is the standard ridge-regression identity $\min_{\alpha} \|r - V^{\top} \alpha\|^2 + \lambda \|\alpha\|^2 = \lambda r^{\top} (V^{\top} V + \lambda I)^{-1} r = r^{\top} (I + V^{\top} V / \lambda)^{-1} r$ written in the singular basis of V ; we claim no novelty for it. In this form h_{λ} is a regularised local Mahalanobis distance from q to the neighbourhood centroid, and the orthogonal term O_{ρ}^2 is present only when the neighbourhood does not span $\mathbb{R}^d$.

Remark 3.4. Proposition 3.3 says that HKNN already combines the two distances of Definition 2.2, with an anisotropic weighting of the tangential coordinates that penalises low-variance directions more ($\lambda/(s_j^2 + \lambda)$ is larger when s_j is small). The model TO combines $\log T_m$ and $\log O_m$ with two scalar weights and a cut-off m ; neither family contains the other. LCD is close to the $\lambda \rightarrow \infty$ end. What the decomposition can add over this family is therefore, in the first place, the depth.

Proposition 3.5 (feature lines). $\text{dist}(q, \text{aff } N_k^c(q)) \leq \ell(q, c) \leq \min_{x_i \in N_k^c(q)} \|q - x_i\|$.

Proof. Every line through two points of $N_k^c(q)$ lies in $\text{aff } N_k^c(q)$, which gives the left inequality; the distance to a line is at most the distance to any of its points, which gives the right one. $\square$

Proposition 3.6 (the depth is not a function of the two distances). *There are neighbourhoods and pairs of queries with equal (T_m, O_m) and different D , and pairs with equal D and different (T_m, O_m) .*

Proof. Take $d = 2$, $N_k^c(q) = \{(-1, 0.3), (0, -0.4), (3, 0.1)\}$, $k = 3$, so $\mu = (2/3, 0)$; the centred scatter matrix is diagonal, $u_1 = e_x$, $u_2 = e_y$, $s = (2.944, 0.510)$, and take $m = 1$. The queries $q_1 = (8/3, 1/2)$ and $q_2 = (-4/3, 1/2)$ both have $T_1 = 2$ and $O_1 = 1/2$, but $D(q_1) = 0.422$ and $D(q_2) = 0.025$ (the unit vectors $S(x_i - q_1)$ point to both sides of q_1 , those of q_2 all point to the right; values to three decimals). For the converse take $d = 1$, $N_k^c(q) = \{-1, 0, 3\}$, $\mu = 2/3$, $m = \rho = 1$: in one dimension $D(q) = 1 - |\frac{1}{k} \sum_i \text{sign}(x_i - q)|$, so $q_3 = 1$ and $q_4 = 5/3$ both have $D = \frac{2}{3}$ (two neighbours below, one above) while $T_1 = \frac{1}{3}$ and $T_1 = 1$. $\square$

Propositions 3.1–3.6 are elementary [proved here]. They fix expectations: relative to HKNN and LPH, the only component that is not already in play is the depth, and Proposition 3.6 shows that it is formally free to help. Whether it does is the empirical question.

4 Protocol and predefined criterion

Datasets. Four standard sets from scikit-learn [23]: iris [10] ($n = 150$, $d = 4$, $C = 3$), wine [1, 16] (178, 13, 3), Wisconsin diagnostic breast cancer [25] (569, 30, 2) and a stratified subsample of the digits (1000 images, 100 per class, $d = 64$). Four synthetic manifolds with $n = 600$: a swiss roll (`make_swiss_roll`, noise 1.0) with three classes given by the terciles of the roll parameter, so that class boundaries run *along* the manifold; two moons (noise 0.3) in the plane; the same moons with ten independent $N(0, 1)$ coordinates appended ($d = 12$), the off-manifold-noise regime; and two nested spheres in $\mathbb{R}^3$ (radii 1 and 1.5, 300 points each, isotropic Gaussian noise of standard deviation 0.2). The digits subsample, the noise coordinates and the spheres are drawn once with the master seed.

Evaluation. 5×5 repeated stratified k -fold cross-validation (25 folds, identical for all methods). Within each fold, coordinates are standardised with training statistics. Every hyper-parameter is chosen on the training fold by leave-one-out accuracy over a fixed grid, ties going to the first grid point: $k_{\text{NN}} \in \{1, 3, 5, 7, 9, 11, 15, 21, 31, 41, 61, 81, 121\}$ (below n_{train}) for kNN; $k \in \{5, 10, 20, 30, 50, 75\}$ for all class-neighbourhood methods, cut per dataset as described below; $m \in \{1, 2, 3, 5, 8, 13, 21\}$ restricted to $m \leq \min(k - 1, d - 1)$ for LPH, T and the logit models; $\lambda = \lambda_{\text{rel}} \bar{s}^2$ with $\lambda_{\text{rel}} \in \{0, 0.1, 1, 10, 100, 10^3, 10^4\}$ for HKNN, where $\bar{s}^2$ is the median over training points of the mean squared singular value of their own-class neighbourhood. Leave-one-out features of the training points (each point excluded from its own class neighbourhood) also serve to fit the logit weights; for the logit models, (k, m) is therefore chosen by the resubstitution accuracy of a logit fitted on those leave-one-out features (the three weights see the points they score), an optimism shared by TOD and its ablations. The resulting optimism (training-fold selection accuracy minus test accuracy) ranges over datasets from 0.4 to 2.2 points for TOD and from 0.5 to 2.0 for HKNN, so it does not tilt the comparison. The test metrics are accuracy and *selective accuracy at 80% coverage* [5]: accuracy on the 80% of test points with the highest confidence, the confidence being the vote fraction for kNN, the relative margin $(\sigma_{(2)} - \sigma_{(1)})/(\sigma_{(1)} + \sigma_{(2)})$ between the two best scores for the argmin classifiers and the maximal posterior for the logit models.

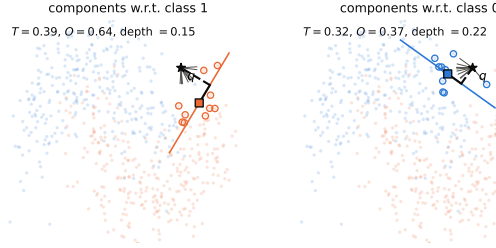


Figure 1: The decomposition on the two-moons data for one query q (star) with respect to each class: circles are the $k = 10$ class neighbours, the square their centroid, the line the one-dimensional tangent subspace T_1 ; the solid segment is the tangential component, the dashed segment the orthogonal one; the short rays are the unit vectors $S(x_i - q)$ whose mean defines the spatial depth.

Statistics. For two methods, the paired fold-level differences in accuracy ($J = 25$) are summarised by their mean and a percentile bootstrap 95% interval with 10 000 resamples [9]; one seeded matrix of resampling indices serves every comparison, so a pair of methods has a single interval. Folds of repeated cross-validation share data, so this interval is too narrow [8]. For the same differences we therefore also report the 95% interval and the two-sided p -value of the corrected resampled t -test of Nadeau and Bengio [21], with the variance inflated by $1/J + n_{\text{test}}/n_{\text{train}} = 1/J + 1/4$ as used for repeated k -fold by Bouckaert and Frank [2]; this correction is known to be conservative, so the truth lies between the two readings, and every count of significant differences below is given in both (bootstrap / NB). The predefined criterion uses the bootstrap interval, the *less* demanding of the two for a positive result. Datasets are not pooled into one test; the criterion counts them, and as a complementary reading across datasets we report the Wilcoxon signed-rank test on the per-dataset mean differences recommended by Demšar [7].

Criterion and preregistration. The criterion was fixed before the first complete run, recorded then only as a constant in the code. The grids above are those of a third complete run: with the earlier grids ($k \leq 30$, $k_{\text{NN}} \leq 41$, $\lambda_{\text{rel}} \leq 100$, $m \leq 8$; run kept in `results/v02/`) the reduced models sat at the top of the k grid in most folds on some datasets, so grids, tie rule and reporting rules were written into a dated preregistration (`PREREGISTRO_SPD001_ronda2.md`, SHA-256 prefix `901e19ff414c`) before the run, not blind to the earlier results. For each dataset the *best reference* is the one of kNN, NFL, HKNN, LPH, LCD, SD with the highest mean accuracy, computed exactly from the integer counts of correct test predictions; references that tie exactly are all kept, and a win (or a loss) must then hold against each of them. **C1:** the decomposition “adds information” if and only if TOD beats the best reference on at least 5 of the 8 datasets with a bootstrap interval strictly above zero. Choosing the best reference after the fact biases the comparison against TOD; the comparison against each reference is also reported. **C2** (attribution): the same rule applied to TOD versus each ablation; an effect supports the attribution only if neither model sits at a truncating top of its k grid in more than half of the folds (preregistered). **C3** (secondary): the same rule on selective accuracy. **E2** (regime): the moons with $p \in \{0, 2, 5, 10, 20\}$ appended noise coordinates; $p = 0$ and $p = 10$ are the moons and the moons with noise of E1.

Grids and residual saturation. The neighbourhood grid is cut at $k_{\text{cap}} = \min_c(n_c - \lceil n_c/5 \rceil) - 1$, the whole class but the point in every training fold, and k_{cap} itself is added when it is below 75 (iris $k \leq 39$, wine $k \leq 37$); on digits $k \leq 50$, a cap fixed in advance for the CPU budget ($d = 64$). With these grids the selected k sits at a truncating top ($k = 75$, or 50 on digits) in more than half of the folds only on the moons with noise (moons+10 noise: NFL 56%, SD 56%, TD 76%); on breast cancer, digits, swiss roll, moons, spheres the largest such fraction over all ten neighbourhood methods is 40% (TD on digits), and on iris and wine the top is the whole class. In E2 the same happens at $p \geq 5$ ($p = 5$: SD 60%, TD 52%; $p = 20$: HKNN 56%, LPH 52%, LCD 52%, SD 92%, TOD 64%, TO 56%, TD 92%, OD 64%). $\lambda_{\text{rel}} = 10^4$ is never chosen, k_{NN} is at its top in at most 4% of the folds in E1, and $m = 21$ (below $\min(k - 1, d - 1)$) in at most 16% for LPH and the logit models (all folds for T on digits); details in `results/tables_appendix.md`.

5 Results

Table 1 gives the mean cross-validated accuracy of every method, Table 2 the primary comparison together with the comparison against HKNN, and Table 3 the ablations; Figure 2 shows the last two

dataset	n	d	C	references						decomposition and ablations				
				kNN	NFL	HKNN	LPH	LCD	SD	TOD	TO	TD	OD	T
iris	150	4	3	94.9	95.2	95.6	96.1	95.7	93.9	96.0	95.6	93.6	96.1	93.6
wine	178	13	3	96.3	96.7	97.3	95.6	97.2	96.4	97.3	97.7	96.4	97.3	90.0
breast cancer	569	30	2	96.6	96.0	96.8	96.7	96.9	95.5	96.6	96.9	96.3	96.4	96.6
digits	1000	64	10	96.4	97.6	97.4	97.4	96.9	91.0	97.2	97.3	92.1	97.1	72.5
swiss roll	600	3	3	96.8	92.9	97.4	97.0	96.9	95.4	97.2	97.2	95.2	97.1	78.9
moons	600	2	2	89.9	80.0	89.8	88.9	89.8	88.9	89.1	89.1	88.1	89.1	78.2
moons+10 noise	600	12	2	83.2	78.2	82.9	83.4	83.1	82.8	83.1	83.0	83.4	82.7	62.8
spheres	600	3	2	82.5	78.6	85.4	84.8	83.5	81.9	85.7	85.2	82.2	85.8	70.8

Table 1: Mean test accuracy (%) over the 25 folds. The best reference on each dataset is in bold (all of them in case of an exact tie; there was none). TOD = tangential + orthogonal + depth; TO, TD, OD drop one component; T, LPH (=O) and SD (=D) are the single components.

dataset	best ref.	TOD – best [boot. CI]	NB CI	NB p	W/L/T	TOD – HKNN [boot. CI]
iris	LPH	−0.13 [−1.07, +0.80]	[−2.86, +2.59]	0.920	5/5/15	+0.40 [−0.27, +1.20]
wine	HKNN	−0.003 [−1.25, +1.36]	[−3.86, +3.85]	0.999	5/8/12	−0.003 [−1.25, +1.36]
breast cancer	LCD	−0.25 [−0.71, +0.18]	[−1.56, +1.06]	0.700	5/8/12	−0.18 [−0.63, +0.28]
digits	NFL	−0.36 [−0.68, −0.06]	[−1.25, +0.53]	0.410	5/13/7	−0.16 [−0.40, +0.08]
swiss roll	HKNN	−0.20 [−0.60, +0.20]	[−1.34, +0.94]	0.721	7/9/9	−0.20 [−0.60, +0.20]
moons	kNN	−0.77 [−1.60, +0.20]	[−3.37, +1.84]	0.549	6/16/3	−0.70 [−1.37, −0.03]
moons+10 noise	LPH	−0.33 [−1.17, +0.47]	[−2.72, +2.06]	0.776	9/12/4	+0.23 [−0.67, +1.13]
spheres	HKNN	+0.33 [−0.43, +1.10]	[−1.87, +2.54]	0.758	12/10/3	+0.33 [−0.43, +1.10]

Table 2: Primary comparison (C1), in accuracy points: mean and percentile bootstrap interval over the 25 folds, interval and p -value with the Nadeau–Bengio (NB) variance, folds won/lost/tied by TOD, and the same difference against HKNN (identical to the first column where HKNN is the best reference: one set of resamples serves every comparison). Significant wins under C1: 0 of 8 (5 required; 0 with NB); significant losses: 1 with the bootstrap (digits), 0 with NB.

graphically. The comparison against every reference, the selective accuracies, the regime scan and the saturation of every grid are tabulated in `results/tables_appendix.md`, generated by the same script as this document; the changes with respect to the narrower grids are in Appendix B. The whole experiment, including the regime scan, took 287 CPU-seconds.

C1: the decomposition against the best reference. The best reference was a local-support method (HKNN, LPH or NFL) on 6 of the 8 datasets, kNN on the moons and LCD on breast cancer (Table 2). TOD beat it with a bootstrap interval above zero on 0 of 8 datasets, against 5 required; the criterion is not met. The mean difference ranged from -0.8 to $+0.3$ points (positive on 1: spheres) and was within one point on 8 of 8 datasets; its bootstrap interval lay entirely below zero on 1 (digits, -0.36 $[-0.68, -0.06]$), which does not survive the corrected t -test ($p = 0.410$; NB interval $[-1.25, +0.53]$). The loss on the moons reported with the narrower grids is gone: there the best reference was an exact tie (kNN/HKNN/LCD, 2697 of 3000 correct) resolved by floating-point rounding, and against kNN the interval already contained zero ($[-1.6, +0.2]$); now kNN is the unique best and $TOD - kNN = -0.77$ $[-1.60, +0.20]$. The largest upper interval limit over all datasets is $+1.4$ points with the bootstrap and $+3.9$ points with the Nadeau–Bengio variance (both on wine): gains larger than the latter are excluded at the 95% level under the cautious reading, gains between the two values only under the anti-conservative one, and smaller gains are not excluded by this study. Across datasets [7], the Wilcoxon signed-rank test on the 8 mean differences gives $W = 5$, two-sided $p = 0.078$ (one-sided $p = 0.97$ for TOD better), and TOD is ahead on 1 of 8.

C2: where the information is. Against HKNN, the reference closest in spirit, the difference ranged from -0.7 to $+0.4$ points, with 0 bootstrap intervals above zero and 1 below (NB: 0 and 0; Table 2); against LPH from -0.3 to $+1.7$ (2 bootstrap intervals above zero, wine, spheres; 0 with NB). On the spheres the depth adds $+0.5$ points over TO (bootstrap interval $[-0.03, +1.0]$, not significant), and HKNN closes the gap to LPH with its penalty: $TOD - HKNN$ is $+0.3$ points there. TOD beat kNN with a bootstrap interval above zero on 2 datasets (0 with NB) and SD on 5 (2 with NB), as the local-distance classifiers do: known advantages of local hyperplane distances over points [28] and of distances over depth alone, not of the decomposition. The accuracies of HKNN, LPH, TOD, TO and OD never differ by more

dataset	– depth (vs TO)	– orthogonal (vs TD)	NB 95% CI (vs TD)	– tangential (vs OD)
iris	+0.40 [−0.53, +1.47]	+2.40 [+1.20, +3.60]	[−1.07, +5.87]	−0.13 [−1.20, +0.93]
wine	−0.46 [−1.48, +0.56]	+0.88 [−0.48, +2.32]	[−3.14, +4.91]	−0.01 [−1.14, +1.13]
breast cancer	−0.25 [−0.70, +0.24]	+0.35 [−0.14, +0.84]	[−1.08, +1.79]	+0.18 [−0.21, +0.56]
digits	−0.14 [−0.32, +0.04]	+5.06 [+4.46, +5.72]	[+3.21, +6.91]	+0.10 [−0.08, +0.28]
swiss roll	−0.07 [−0.40, +0.27]	+1.93 [+1.20, +2.70]	[−0.23, +4.10]	+0.07 [−0.37, +0.57]
moons	+0.03 [−0.40, +0.43]	+1.03 [+0.03, +2.00]	[−1.79, +3.85]	+0.03 [−0.37, +0.47]
moons+10 noise	+0.10 [−0.70, +0.83]	−0.27 [−1.17, +0.60]	[−2.86, +2.32]	+0.40 [−0.73, +1.43]
spheres	+0.47 [−0.03, +1.00]	+3.53 [+2.23, +4.70]	[+0.01, +7.05]	−0.10 [−0.87, +0.63]

Table 3: Ablations (C2): TOD minus the model with the named component removed, in accuracy points, mean and bootstrap 95% interval; for the orthogonal component also the interval with the Nadeau–Bengio variance. A positive value means the component helped. Both intervals for all three ablations are in `results/tables_appendix.md`.

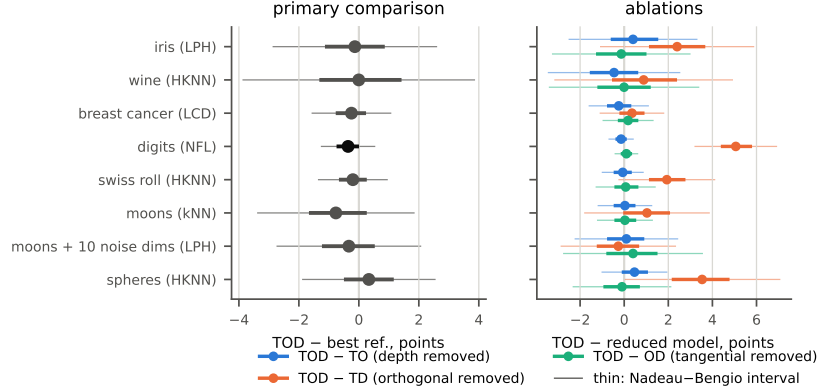


Figure 2: Left: TOD minus the best reference per dataset (C1). Right: the three ablations (C2). Thick segments: bootstrap 95% intervals; thin segments: intervals with the Nadeau–Bengio variance; black (left) marks a bootstrap interval excluding zero.

than 2.1 points on any dataset (the largest spread on wine): they behave as one family, as Proposition 3.3 and Remark 3.4 suggest.

The ablations (Table 3, Figure 2) say where the information sits. Removing the orthogonal component costs between -0.3 and $+5.1$ points, with a bootstrap interval above zero on 5 of 8 datasets (iris, digits, swiss roll, moons, spheres) and a Nadeau–Bengio interval above zero on 2 (digits, spheres). By the preregistered rule all 5 effects are robust to the grid: neither TOD nor TD sits at a truncating top of the k grid in more than half of the folds (at most 40%, TD on digits). Two effects seen with the narrower grids did not survive (Table 5): on the moons with noise TOD – TD went from $+2.7$ to -0.3 points because TD needed larger neighbourhoods (at $k = 30$ in 100% of the folds before, still at $k = 75$ in 76% now), so the orthogonal contribution there is not established; on breast cancer it went from $+1.1$ to $+0.4$, an interval now containing zero. The attribution rests on 5 datasets (bootstrap; 2 with NB) and on the fitted weights: the weight on $\log O$ is the largest on 8 of 8 datasets (between 1.53 and 3.89), that on $\log T$ lies between 0.17 and 1.48 and that on $\log E$ between -1.06 and 1.95, negative on 3 datasets, so its sign is not even stable. Removing the depth changes accuracy by -0.5 to $+0.5$ points (0 bootstrap intervals above zero, 0 below; 0 and 0 with NB), removing the tangential component by -0.1 to $+0.4$ (0 and 0; 0 and 0 with NB) (largest upper limits $+1.5$ and $+1.4$ points; NB $+3.3$ and $+3.5$). The tangential distance alone (T , from 62.8% on moons+10 noise to 96.6%) is the least accurate method on 6 of 8 datasets, consistent with it being a within-support coordinate; it approaches the centroid distance only where m can approach d .

C3: abstention. With each method abstaining on its own least confident fifth of the test points, TOD differed from the best reference of Table 2 by -2.3 to $+0.3$ points; the bootstrap interval lay above zero on 1 dataset (digits, $+0.2$ points; NB $[-0.2, +0.6]$) and below zero on 1 (moons, -2.3); with NB, 0 above and 0 below. Against the reference that is best *by selective accuracy*, the differences run from -3.2 to $+0.1$ points with 0 bootstrap intervals above zero and 2 below (moons, moons+10 noise); on digits that reference is LCD and the difference is $+0.1$ $[-0.1, +0.3]$. The posterior of the three-component logit is therefore not a better abstention signal than the margins of the reference scores, and the depth does not improve selective accuracy here.

claim	status
C1: TOD beats the best reference on 0 of 8 datasets (5 required; 0 with NB); the decomposition does not add information in the predefined sense	verified (negative)
Upper 95% limit of the gain over the best reference: at most +1.4 points (bootstrap) / +3.9 points (NB) on every dataset	verified
C2: removing the orthogonal component costs up to +5.1 points, interval above zero on 5/8 (bootstrap; iris, digits, swiss roll, moons, spheres), 2/8 (NB), all robust to the widened grids; depth and tangential components removable without significant loss (0/8 and 0/8 bootstrap, 0/8 and 0/8 NB)	verified
C2 on breast cancer, moons+10 noise: the orthogonal effects of the narrower grids do not survive; on the moons with noise the reduced model is still grid-limited	not established
C3: no gain in selective accuracy at 80% coverage (1 bootstrap interval above zero against the best reference by accuracy, 0 against the best by selective accuracy, 0 with NB)	verified
E2: no off-manifold-noise regime with $p \leq 20$ in which TOD exceeds the best reference; the depth helps the logit model only at $p = 2$ (+0.9 points, bootstrap; NB $p = 0.282$); orthogonal contribution for $p \geq 5$ grid-limited	verified (partly)
HKNN, LPH, TOD, TO, OD differ by at most 2.1 points on every dataset	verified
Gains below +3.9 points (below +1.4 under the bootstrap); behaviour on large or high-dimensional data; other depths	not established

Table 4: Empirical claims and their status; the propositions of Section 3 are labelled where they are stated. “Verified” means produced by the scripts of Appendix A with the fixed seed on the datasets of Section 4; it is not a general statement.

E2: the off-manifold-noise regime. Appending noise coordinates to the moons lowers every method (`results/tables_appendix.md`; from $p = 0$ to $p = 20$, kNN by 7.5 points, TOD by 6.9). The best reference at each p was one of {HKNN, LCD, LPH, kNN}; TOD minus the best reference ranged from -0.8 to -0.1 points with 0 bootstrap intervals above zero and 1 below ($p = 20$; 0 and 0 with NB). The depth contribution ($\text{TOD} - \text{TO}$) ranged from $+0.03$ to $+0.9$ points and the tangential one ($\text{TOD} - \text{OD}$) from $+0.03$ to $+0.4$, with 1 and 0 bootstrap intervals above zero. The orthogonal contribution ($\text{TOD} - \text{TD}$) ranged from -0.3 to $+2.4$, with an interval above zero only at $p = 0$, $p = 2$; at $p = 5$ and $p = 20$ it fell from $+2.1$ and $+2.3$ with the narrower grids to $+0.9$ and $+0.03$, while TD and SD still choose $k = 75$ in most folds: with five or more noise coordinates the comparison is limited by the grid and the orthogonal contribution is not established. The single bootstrap interval above zero for the depth in the whole study is at $p = 2$ ($+0.9$ points [$+0.3$, $+1.5$]; NB [-0.8 , $+2.6$], $p = 0.282$). It is one of 11 distinct depth intervals in E1 and E2 ($p = 0 = \text{moons}$, $p = 10 = \text{moons}+10 \text{ noise}$), compatible with chance, and there TOD remains -0.1 points from the best reference (LCD, 87.9%): the depth helps the logit model catch up, not overtake. In no noise regime in this range does the depth or the tangential component add to the best reference.

6 What can be claimed

Table 4 lists the claims with their status. The answer to the line’s question, in the operational sense fixed in Section 4, is negative: the tangential–orthogonal–depth decomposition does not add predictive information beyond HKNN-type local distances, which are a regularised hull distance where $d \geq k$ and a regularised local Mahalanobis distance to the centroid where $k > d$ (Remark 3.2, the case of most of our datasets). HKNN already contains the tangential part with a sensible weighting (Proposition 3.3). The depth was formally free to help (Proposition 3.6) and did not add to the best reference, in accuracy or in abstention, on any of the 8 datasets or in the noise regime scanned; its one bootstrap-significant contribution ($p = 2$ in E2, not significant with the Nadeau–Bengio variance) brought the logit model level with the references, not past them. Widening the grids left C1 unchanged and removed two of the seven orthogonal effects of the previous version.

Limitations. The datasets are small and, for 5 of them (iris, wine, breast cancer, digits, swiss roll), the best reference is at or above 96% accuracy, close to the ceiling; one test error is worth 3.33 accuracy points on iris and 2.78 on wine, so “within one point” is there below the resolution of a single fold and most folds are exact ties (W/L/T in Table 2). The study has power to exclude gains of a few points, +3.9 with the Nadeau–Bengio variance and +1.4 with the fold-level bootstrap, which is anti-conservative

because the folds of repeated cross-validation share data; it does not exclude smaller gains. Every count of significant differences is given in both readings. The logit weights were fitted on the leave-one-out features used for selection (optimism 0.4–2.2 points for TOD, 0.5–2.0 for HKNN). The grids were widened after seeing the earlier runs, under a preregistration that was not blind to them; on the moons with noise and in E2 with $p \geq 5$ several methods, TD among them, still choose the largest k in most folds, and on digits k was capped at 50 for cost. Only one depth and one combination rule (a conditional logit without intercepts) were tried; the negative finding is about the tested instance. The tangent dimension m was tuned globally, and the score-based methods use no class priors.

Next steps for the research line. The line’s stated objective is discharged with a negative answer and can be closed. If it were reopened, the directions the present results leave open are: (a) depths with a different mechanism (halfspace or simplicial depth on the neighbourhood, or the symmetrised local depth of 22) under the same criterion; (b) the decomposition as a *descriptive* tool for HKNN, e.g. choosing λ from the spectrum (s_j) via Proposition 3.3 rather than by search, a question about HKNN and not about a new classifier; (c) larger and higher-dimensional data, where the hull term does not vanish (Remark 3.2); (d) on the moons with noise and in E2, neighbourhoods up to the whole class ($k_{\text{cap}} = 239$), and $k = 75$ on digits, to settle the orthogonal effects that remain grid-limited.

A Reproducibility

The directory contains `experiments/support_geometry.py` (the whole experiment, E1 and E2; flag `-fast` for a reduced run), `experiments/make_figures.py` and `experiments/make_numbers.py`, which converts `results/results.json` and `results/results_regime.json` into the macros and table bodies used by this document and into `results/tables_appendix.md`, so that no number in the text is typed; it also recomputes the Nadeau–Bengio intervals, saturation fractions, modal hyper-parameters and the comparison with the narrower grids (`results/v02/`) from the per-fold lists. The run used seed 20260930, Python 3.11.15, NumPy 2.4.6 [13], SciPy 1.17.1 [29] and scikit-learn 1.9.1 [23], single-threaded BLAS, and took 296 s wall / 287 s CPU on a shared four-core machine. Computation: per fold, all squared distances are computed once; for each class the k_{max} nearest same-class points are found by partial sorting, the feature-line distances, depths and mean distances for every k come from prefix minima and cumulative sums over them, and each neighbourhood is decomposed through the eigen-decomposition of its smaller Gram matrix, directions with $s_j^2 < 10^{-10} s_1^2$ being treated as null so that the hull distance is exact also for non-generic neighbourhoods (constant pixels in digits); the test fold is processed only at the selected k . With the earlier grids this code reproduces the earlier run fold by fold on the moons.

B The effect of the grids

The modal hyper-parameters, the fitted weights and the per-method saturation of every grid are tabulated in `results/tables_appendix.md`; Table 5 summarises what changed when the grids were widened.

dataset	TOD – TD, grids $k \leq 30$		TOD – TD, preregistered grids		TOD – best reference	
	[boot. CI]	% top	[boot. CI]	% top	$k \leq 30$	preregistered
iris	+2.5 [+1.3, +3.7]	12	+2.4 [+1.2, +3.6]	0 ^c	−0.3 [−1.2, +0.7]	−0.1 [−1.1, +0.8]
wine	+0.7 [−0.6, +2.0]	8	+0.9 [−0.5, +2.3]	0 ^c	+0.1 [−1.0, +1.3]	−0.003 [−1.3, +1.4]
breast cancer	+1.1 [+0.5, +1.7]	16	+0.4 [−0.1, +0.8]	0	−0.2 [−0.7, +0.2]	−0.2 [−0.7, +0.2]
digits	+5.1 [+4.4, +5.8]	52	+5.1 [+4.5, +5.7]	40	−0.3 [−0.6, −0.1]	−0.4 [−0.7, −0.1]
swiss roll	+1.9 [+1.2, +2.7]	0	+1.9 [+1.2, +2.7]	0	−0.2 [−0.6, +0.2]	−0.2 [−0.6, +0.2]
moons	+1.1 [+0.2, +2.0]	76	+1.0 [+0.03, +2.0]	16	−0.8 [−1.6, +0.2]	−0.8 [−1.6, +0.2]
moons+10 noise	+2.7 [+1.5, +3.8]	100	−0.3 [−1.2, +0.6]	76	−0.4 [−1.3, +0.6]	−0.3 [−1.2, +0.5]
spheres	+3.5 [+2.2, +4.7]	0	+3.5 [+2.2, +4.7]	0	+0.2 [−0.5, +1.0]	+0.3 [−0.4, +1.1]
E2, $p = 2$	+2.3 [+1.7, +2.9]	64	+2.4 [+1.8, +3.0]	0	−0.1 [−0.8, +0.6]	−0.1 [−0.8, +0.6]
E2, $p = 5$	+2.1 [+1.3, +3.1]	80	+0.9 [−0.03, +1.8]	52	−0.3 [−1.0, +0.3]	−0.5 [−1.4, +0.3]
E2, $p = 20$	+2.3 [+0.9, +3.7]	100	+0.03 [−0.6, +0.6]	92	−0.4 [−1.2, +0.3]	−0.6 [−1.1, −0.1]

Table 5: Effect of widening the grids: the orthogonal ablation and the primary comparison with the earlier grids ($k \leq 30$; run kept in `results/v02/`) and with the preregistered ones, both summarised with the same bootstrap resamples and the tie rule of Section 4; “% top”: folds in which TD chose the largest k of its grid.

References

- [1] Stefan Aeberhard and M. Forina. Wine. UCI Machine Learning Repository, 1991.
- [2] Remco R. Bouckaert and Eibe Frank. Evaluating the replicability of significance tests for comparing learning algorithms. In *Advances in Knowledge Discovery and Data Mining (PAKDD 2004)*, volume 3056 of *Lecture Notes in Computer Science*, pages 3–12, Berlin, 2004. Springer. doi: 10.1007/978-3-540-24775-3_3.
- [3] Hakan Cevikalp, Bill Triggs, and Robi Polikar. Nearest hyperdisk methods for high-dimensional classification. In William W. Cohen, Andrew McCallum, and Sam T. Roweis, editors, *Proceedings of the 25th International Conference on Machine Learning (ICML)*, pages 120–127, Helsinki, 2008. Omnipress.
- [4] Probal Chaudhuri. On a geometric notion of quantiles for multivariate data. *Journal of the American Statistical Association*, 91(434):862–872, 1996.
- [5] C. K. Chow. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1): 41–46, 1970.
- [6] Thomas M. Cover and Peter E. Hart. Nearest neighbor pattern classification. *IEEE Transactions on Information Theory*, 13(1):21–27, 1967.
- [7] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7: 1–30, 2006. URL <https://jmlr.org/papers/v7/demsar06a.html>. JMLR assigns no DOI.
- [8] Thomas G. Dietterich. Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation*, 10(7):1895–1923, 1998.
- [9] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, New York, 1993.
- [10] Ronald A. Fisher. The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, 7(2):179–188, 1936.
- [11] Evelyn Fix and Joseph L. Hodges. Discriminatory analysis. Nonparametric discrimination: consistency properties. *International Statistical Review*, 57(3):238–247, 1989. doi: 10.2307/1403797. Reprint of USAF School of Aviation Medicine Report 4, Randolph Field, Texas, 1951.
- [12] Anil K. Ghosh and Probal Chaudhuri. On maximum depth and related classifiers. *Scandinavian Journal of Statistics*, 32(2):327–350, 2005.
- [13] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, et al. Array programming with NumPy. *Nature*, 585: 357–362, 2020.
- [14] Trevor Hastie and Robert Tibshirani. Discriminant adaptive nearest neighbor classification. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 18(6):607–616, 1996.
- [15] Nandakishore Kambhata and Todd K. Leen. Dimension reduction by local principal component analysis. *Neural Computation*, 9(7):1493–1516, 1997. doi: 10.1162/neco.1997.9.7.1493.
- [16] Markelle Kelly, Rachel Longjohn, and Kolby Nottingham. The UCI machine learning repository. <https://archive.ics.uci.edu>, 2023. The repository as a whole has no DOI; individual datasets do.
- [17] Jun Li, Juan A. Cuesta-Albertos, and Regina Y. Liu. DD-classifier: nonparametric classification procedure based on DD-plot. *Journal of the American Statistical Association*, 107(498):737–753, 2012.
- [18] Stan Z. Li and Juwei Lu. Face recognition using the nearest feature line method. *IEEE Transactions on Neural Networks*, 10(2):439–443, 1999.
- [19] Regina Y. Liu. On a notion of data depth based on random simplices. *The Annals of Statistics*, 18(1):405–414, 1990.
- [20] Philip Milasevic and Gilles R. Ducharme. Uniqueness of the spatial median. *The Annals of Statistics*, 15(3):1332–1333, 1987. doi: 10.1214/aos/1176350511.
- [21] Claude Nadeau and Yoshua Bengio. Inference for the generalization error. *Machine Learning*, 52(3):239–281, 2003.
- [22] Davy Paindaveine and Germain Van Bever. From depth to local depth: a focus on centrality. *Journal of the American Statistical Association*, 108(503):1105–1119, 2013. doi: 10.1080/01621459.2013.813390.
- [23] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [24] Robert Serfling. A depth function and a scale curve based on spatial quantiles. In Yadolah Dodge, editor, *Statistical Data Analysis Based on the L_1 -Norm and Related Methods*, Statistics for Industry and Technology, pages 25–38. Birkhäuser, Basel, 2002.
- [25] W. Nick Street, William H. Wolberg, and Olvi L. Mangasarian. Nuclear feature extraction for breast tumor diagnosis. In *Biomedical Image Processing and Biomedical Visualization (IS&T/SPIE 1993)*, volume 1905 of *Proceedings of SPIE*, pages 861–870, 1993.
- [26] John W. Tukey. Mathematics and the picturing of data. In *Proceedings of the International Congress of Mathematicians (Vancouver, 1974)*, volume 2, pages 523–531, 1975.
- [27] Yehuda Vardi and Cun-Hui Zhang. The multivariate L_1 -median and associated data depth. *Proceedings of the National Academy of Sciences*, 97(4):1423–1426, 2000. doi: 10.1073/pnas.97.4.1423.
- [28] Pascal Vincent and Yoshua Bengio. K-local hyperplane and convex distance nearest neighbor algorithms. In Thomas G. Dietterich, Suzanna Becker, and Zoubin Ghahramani, editors, *Advances in Neural Information Processing Systems 14*, pages 985–992. MIT Press, Cambridge, MA, 2002.
- [29] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020.
- [30] Yijun Zuo and Robert Serfling. General notions of statistical depth function. *The Annals of Statistics*, 28(2):461–482, 2000.